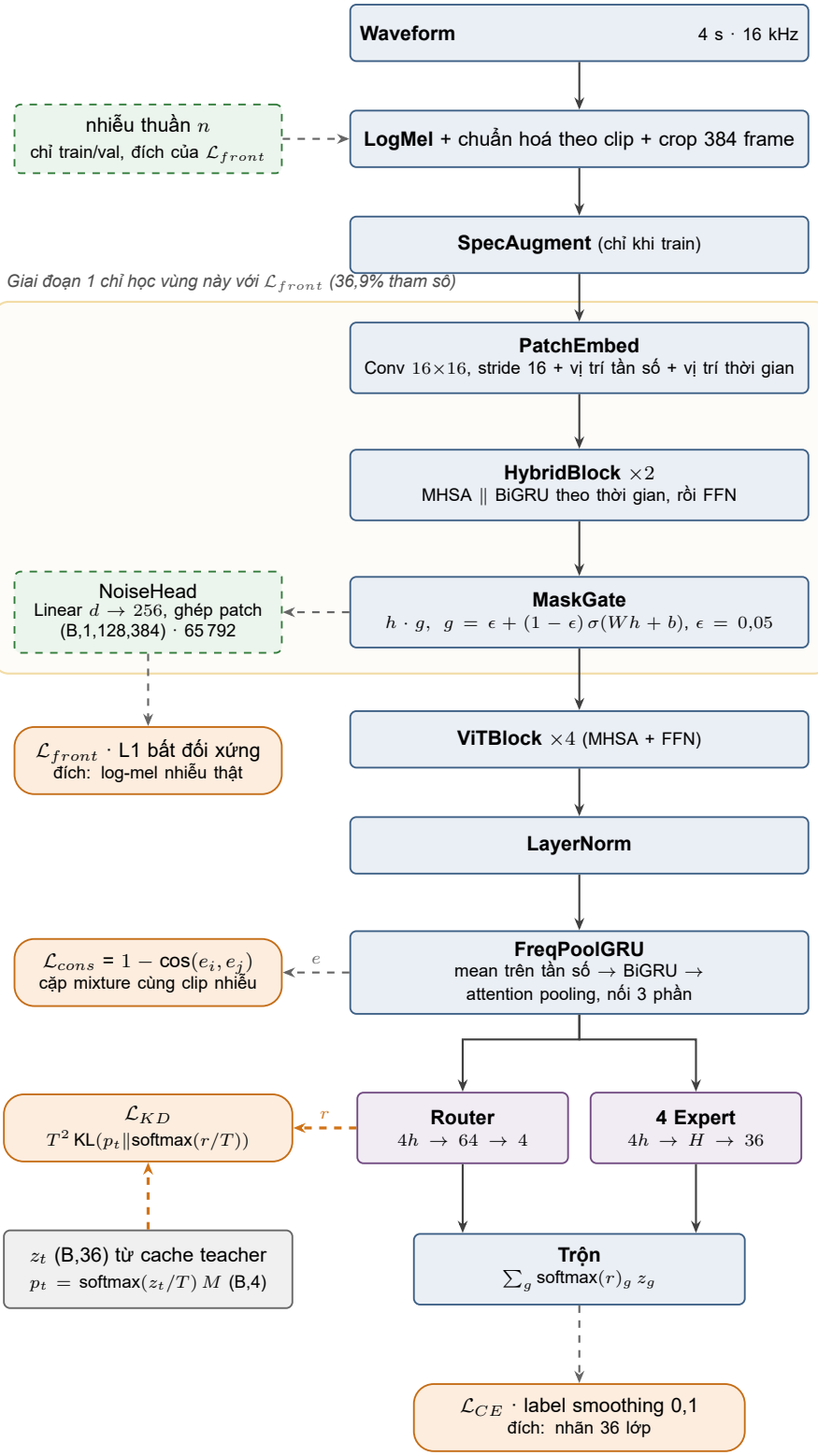


	Shape ra (M: $d = 256$, $h = 128$)	Tham số (M)
Waveform	(B, 64 000)	—
LogMel + chuẩn hoá theo clip + crop 384 frame	(B, 1, 128, 384)	—
SpecAugment (chỉ khi train)	(B, 1, 128, 384)	—
PatchEmbed	(B, 8, 24, 256)	73 984
HybridBlock $\times2$	(B, 8, 24, 256)	2 172 420
MaskGate	(B, 8, 24, 256) và cổng g cùng shape	65 792
ViTBlock $\times4$ (MHSA + FFN)	(B, 8, 24, 256)	3 159 040
LayerNorm	(B, 8, 24, 256)	512
FreqPoolGRU	(B,24,256) → (B,24,256) → e (B, 512)	312 960
Router / 4 Expert	r (B, 4) và z (B, 4, 36)	33 092 + 562 320
Trộn	logits (B, 36)	—

Tổng 6 445 912 (suy luận 6 380 120, bỏ NoiseHead)